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Saveetha Institute of Medical And Technical Sciences
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Title 1: Music Streaming Application Playlist Management Using Doubly Linked List

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INTRODUCTION

- Music streaming applications store songs in playlists for easy access.
- Users can add, remove, and reorder songs at any time.
- The playlist changes dynamically based on user preferences.
- Efficient playlist management improves performance and user experience.
- A **Doubly Linked List** is well suited because it supports flexible insertion, deletion, and navigation between songs.



OPERATIONS



- **Insert:** Add a new song at the beginning, end, or any position in the playlist.
- **Delete:** Remove a selected song from the playlist.
- **Traverse:** Move to the next or previous song easily.
- **Reorder:** Change the position of songs by moving them within the playlist.
- **Display:** Show the current playlist in the desired order.



APPLICATIONS

- **Music streaming platforms** (Spotify, Apple Music, YouTube Music)
- **Media players** for playlist management
- **Audio queue management** (Next/Previous song)
- **Podcast playlist organization**
- **Offline music library management**



CONCLUSION



- Doubly Linked List provides flexible playlist management.
- It supports fast insertion, deletion, and reordering of songs.
- Forward and backward navigation is simple and efficient.
- It handles dynamic playlist updates effectively.
- Therefore, it is the most suitable data structure for music streaming playlists.



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Title 2: Dictionary Application

Word Search Using Hash Table



INTRODUCTION



- A dictionary application stores a large collection of words and meanings.
- Users need to search for words quickly.
- Fast search improves the user experience.
- The data structure should support efficient insertion and deletion.
- A **Hash Table** is the most suitable choice because it provides very fast search performance.



OPERATIONS



- Insert a new word into the dictionary.
- Search for a word.
- Update the meaning of a word.
- Delete an existing word.
- Display the search result.



APPLICATIONS



- Online dictionary applications
- Spell checker software
- Search engines
- Contact lookup systems
- Word suggestion and autocomplete



CONCLUSION



- Hash Tables provide very fast search performance.
- They support quick insertion and deletion of words.
- Average search time is **$O(1)$** .
- They efficiently organize large amounts of data.
- Therefore, a Hash Table is the best data structure for dictionary applications.



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Title 3: Task Scheduling System Using Priority Queue



INTRODUCTION



- A task scheduling system executes jobs based on their priority.
- High-priority tasks are processed before low-priority tasks.
- This improves system performance and resource utilization.
- The scheduling system should manage tasks efficiently.
- A **Priority Queue (Heap)** is the most suitable data structure.



OPERATIONS



- Insert a new task.
- Assign a priority to the task.
- Execute the highest-priority task.
- Remove the completed task.
- Update the priority of a task.



APPLICATIONS



- CPU process scheduling
- Printer job scheduling
- Hospital emergency systems
- Network packet scheduling
- Cloud computing task management



CONCLUSION



- A Priority Queue executes the highest-priority task first.
- It provides efficient insertion and deletion of tasks.
- It improves scheduling speed and system performance.
- It ensures better resource utilization.
- Therefore, a Priority Queue (Heap) is the best data structure for task scheduling systems.